

ECSE 343 Project

Numerical Integration

This project is to be done in groups of 3 to 4 students. Your goal is to develop a multidimensional numerical integration function similar to the MATLAB functions `integral2` or `integral3`.

Numerical integration, particularly multidimensional numerical integration, is an important and challenging problem in engineering. One of the key difficulties is that the function being integrated may be CPU expensive to evaluate. A “function evaluation” may, for example, mean running a long circuit simulation that may take hours. As a result, in general,

- a. We cannot compute the integral using closed-form expressions because (in practical situations, we may not know the integrand).
- b. We want to use a small number of function evaluations as they are computationally expensive.

In this project, you are provided with several integrand functions for testing purposes, but you may test your algorithms on additional problems as well; you need to design an algorithm to compute multidimensional numerical integrals. The integrand functions are given as MATLAB functions (see Appendix).

Suppose the integrand function can be written as $f(x_1, x_2, \dots, x_d)$. We want to evaluate the following definite multidimensional integral,

$$I = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_d}^{b_d} f(x_1, x_2, \dots, x_d) dx_d \dots dx_2 dx_1 \quad (1)$$

The above requires computing the definite integral over d -dimensions. For a given dimension x_i the limits of the integration are $[a_i, b_i]$, i.e., x_i lies between the boundaries a_i and b_i . Your function should numerically evaluate the above integral while meeting the following criteria:

- a. Your function can only evaluate the integrand.
- b. The number of times the integrand is evaluated should be minimized in order to reduce CPU cost.
- c. The best possible accuracy should be obtained for the given number of function evaluations.

There are several approaches to integrate the function over multiple dimensions; one class of methods are Monte-Carlo based methods. Other examples are methods using Quadrature Rules and Sparse Grids.

Your tasks in this project are listed below,

- a. Search the literature to identify and evaluate possible approaches for solving this problem, including the Monte-Carlo based methods.

- b. Implement a Monte-Carlo based approach as well as another approach that you choose based on the project goals above (primarily reducing the number of function evaluations required for a given accuracy).
- c. Test both the Monte-Carlo-based method that you implemented as well as the other approach that you chose to implement. Vary the number of function evaluations and comment on each approach's accuracy. Comment on how the accuracy changes for different numbers of function evaluations.

To compare the accuracy of the proposed approach and the Monte-Carlo you can use the built in MATLAB functions for computing the integrals in 1-D, 2-D, and 3-D.

The names of MATLAB functions that evaluate the value of the integrands are given below:

1. func1.m, integrate this function when the limits of the integral are $[-3 \ 3]$ for 1-D, 2-D, and 3-D cases. This is a d -dimensional Gaussian function given by,

$$f(x_1, x_2, \dots, x_d) = \frac{1}{(2\pi)^{d/2}} e^{0.5 \cdot \sum_{i=1}^d -x_i^2}$$

2. func2.m, integrate this function when the limits of the integral are $[0 \ 1]$ for 1-D, 2-D, and 3-D cases. This function provides the output of the following d -dimensional function,

$$f(x_1, x_2, \dots, x_d) = \prod_{i=1}^d |4x_i - 2|$$

3. func3.m, integrate this function when the limits of the integral are $[0 \ 1]$ for 1-D, 2-D, and 3-D cases. This function has the following d -dimensional function,

$$f(x_1, x_2, \dots, x_d) = \prod_{i=1}^d \frac{1}{(a^{-2} + (x_i - u)^2)} \text{ for } a = 5, u = 0.5$$

4. func4.m, integrate this function when the limits of the integral are $[0 \ 1]$ for 1-D, 2-D, and 3-D cases. This d -dimensional function has the following form,

$$f(x_1, x_2, \dots, x_d) = \cos \left(2\pi u + \sum_{i=1}^d ax_i \right) \text{ for } a = 5, u = 0.5$$

5. func5.m, integrate this function when the limits of the integral are $[0 \ 1]$ for 1-D, 2-D, and 3-D cases. This d -dimensional function can be written as,

$$f(x_1, x_2, \dots, x_d) = \prod_{i=1}^d \frac{|4x_i - 2| + a}{1 + a} \text{ for } a = 5$$

6. func6.m, integrate this function when the limits of the integral are $[0 \ 1]$ for 1-D, 2-D, and 3-D cases. This d -dimensional function can be written as,

$$f(x_1, x_2, \dots, x_d) = \left(1 + \sum_{i=1}^d ax_i \right)^{-(d+1)} \text{ for } a = 5$$

Deliverables

Your deliverables will include two items. The first is a written report (maximum 5 pages, but no minimum as long as you explain the required content). The second is the MATLAB code that you developed. Make sure the code is well commented. Your report should be structured like an IEEE journal paper with an abstract, introduction, body, and conclusion. It must address at least the following points:

1. A description of the theoretical foundation of both methods that you have implemented.
2. A description of the algorithm used for each method in the form of a pseudo-code.
3. A justification for the choice of the second non-Monte-Carlo method.
4. A discussion about the advantages and disadvantages of each method.
5. A discussion of the numerical experiments that were conducted to validate, compare, and evaluate the algorithms used.
6. A description of the contribution of each team member to the project.
7. A final discussion and conclusion.

Grading

The grading scheme is as follows:

- 50% submitted code: Does the code work? Is it efficient? Is it well structured? Is it clear and well commented? The grade will be split evenly for the Monte Carlo approach and the other approach that you choose.
- 50% Project report (see the file “grading rubric” for details on the report grade)

Appendix

Below is an example of the MATLAB function that computes the value, y , of the integrand at the input sample point, $\mathbf{x}$. As stated in the introduction we need to compute the multidimensional integrals, therefore, the input sample $\mathbf{X}$ can be a multidimensional vector containing the values of the samples at different dimension.

```
function [y] = evaluate_integrand(X)  
  
% X is a row vector containing sample values  
% For a d-dimensions, X = [ x1 x2 x3 ... xd].  
  
% y is the value of the function evaluated at X  
% y = f(x1, x2, x3, ... ,xd)  
  
End
```

For example, suppose the integrand function can be written as $f(x_1, x_2, \dots, x_d)$ and we want to compute the following definite multidimensional integral,

$$I = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_d}^{b_d} f(x_1, x_2, \dots, x_d) dx_d \dots dx_2 dx_1$$

Then, the above MATLAB function will provide you the output value, y , of the $f(x_1, x_2, \dots, x_d)$ at the input sample $\mathbf{X} = [x_1 \ x_2 \ \dots \ x_d]$.